

An Explicit Evaluation of a Fibonacci Character Sum for Primes with Full Rank of Apparition

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Abstract

Let $p \geq 7$ be a prime, χ_p the quadratic character mod p , and (F_n) the Fibonacci sequence. The rank of apparition $\alpha(p)$ is the smallest $m \geq 1$ with $p \mid F_m$. Classically $\alpha(p) \mid p - \chi_p(5)$; hence the extremal condition $\alpha(p) = p - 1$ forces $p \equiv \pm 1 \pmod{5}$.

In this full-rank regime we show first that $p \equiv 3 \pmod{4}$ and that the quadratic nonresidue root of $x^2 - x - 1$ in $\mathbb{F}_p$ is a primitive root of $\mathbb{F}_p^\times$. The Fibonacci character sum then collapses to

$$\sum_{n=1}^{p-1} \chi_p(F_n) = \begin{cases} +1, & p \equiv 11 \pmod{20}, \\ -1, & p \equiv 19 \pmod{20}. \end{cases}$$

The proof reduces the sum to a single character value $\chi_p(r_- - r_+)$, where r_- and r_+ denote the nonresidue and residue roots of $x^2 - x - 1$, via a primitive-root substitution; this value is then identified through a discriminant criterion linking the roots of $x^2 - x - 1$ to fifth roots of unity in $\mathbb{F}_p$. The result may be viewed as a Fibonacci analogue of an Artin-type primitivity condition. The theoretical result is accompanied by an independent computational verification up to 2×10^6 .

Keywords: Fibonacci numbers, Lucas sequences, character sums, rank of apparition, primitive roots, fifth roots of unity

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1. Introduction

The Fibonacci sequence $(F_n)_{n \geq 0}$, $F_0 = 0$, $F_1 = 1$, $F_{n+1} = F_n + F_{n-1}$, is the prototypical non-degenerate binary linear recurrence. For a prime $p \neq 5$, its mod- p behaviour is encoded by two classical invariants: the *Pisano period* $\pi(p)$ and the *rank of apparition*

$$\alpha(p) := \min\{m \geq 1 : p \mid F_m\}.$$

A classical result going back to Lucas [1] states

$$\alpha(p) \mid p - \chi_p(5), \tag{1}$$

where χ_p denotes the quadratic character. Hence in the split case $\chi_p(5) = 1$, equivalently $p \equiv \pm 1 \pmod{5}$, the maximal possible value of $\alpha(p)$ is $p - 1$. Throughout this paper, the term *full-rank regime* refers to the extremal split value $\alpha(p) = p - 1$; the hypothesis itself will force the split case.

1.1. The Fibonacci character sum

We study

$$S(p) := \sum_{n=1}^{p-1} \chi_p(F_n).$$

A priori $S(p)$ might be expected to fluctuate as p varies, much like character sums attached to general linear recurrences [2, 3, 4]. Our main theorem shows that in the extremal regime $\alpha(p) = p - 1$ this apparent complexity collapses completely.

Theorem 1.1 (Main Theorem). *Let $p \geq 7$ be a prime with $\alpha(p) = p - 1$. Then $p \equiv 11 \pmod{20}$ or $p \equiv 19 \pmod{20}$, and*

$$S(p) = \begin{cases} +1, & p \equiv 11 \pmod{20}, \\ -1, & p \equiv 19 \pmod{20}. \end{cases}$$

1.2. Why this regime?

The hypothesis $\alpha(p) = p - 1$ has an intrinsic primitive-root interpretation. In the split case, Lemma 3.1 below identifies $\alpha(p)$ with the order of $-r^2$, where r is either root of $x^2 - x - 1$ in $\mathbb{F}_p$. The condition is thus a *Fibonacci analogue* of the kind of primitivity condition studied in Artin's primitive root conjecture.

The non-obvious content of Lemma 3.2 below is that full rank first forces the split case and the congruence $p \equiv 3 \pmod{4}$, and then the primitivity of $-r_-^2$ lifts to the nonresidue root r_- itself. The lifting mechanism is purely group-theoretic: since r_- is a nonresidue, $-1 \in \langle r_- \rangle$, and hence $-r_-^2 \in \langle r_- \rangle$; as $-r_-^2$ already has order $p - 1$, the root r_- must be primitive. This lift is what drives the entire evaluation of $S(p)$.

1.3. Outline of the proof

The proof proceeds in three steps.

Step I. *Primitivity*. We show that $\alpha(p) = p - 1$ forces $x^2 - x - 1$ to split over $\mathbb{F}_p$, forces $p \equiv 3 \pmod{4}$, and makes the nonresidue root r_- a primitive root of $\mathbb{F}_p^\times$ (Lemma 3.2).

Step II. *Structural identity*. Using Binet's formula and Step I, the substitution $u = r_-^n$ converts the index n into a complete traversal of $\mathbb{F}_p^\times$ and yields $S(p) = \chi_p(r_- - r_+)$ (Proposition 5.1).

Step III. *Explicit sign*. A discriminant criterion linking the roots of $x^2 - x - 1$ to fifth roots of unity in $\mathbb{F}_p$ determines $\chi_p(r_- - r_+)$ in closed form (Lemma 6.3).

1.4. Related literature

The arithmetic of Fibonacci/Lucas sequences modulo primes begins with Lucas [1] and Wall [5]; standard treatments are [6, 7, 8, 9]. Distribution of $\alpha(p)$ and related Pisano quantities is studied in [? 10, 11, 12]. Recent investigations on the p -adic and arithmetic structure of Lucas-type sequences include [? ?]. Character sums attached to linear recurrences appear in [2, 3, 4, 13]. The primitive-root perspective connects the full-rank condition with Artin-type problems; see the survey [14] and Hooley's conditional density theorem [15] for the classical setting. To the best of the author's knowledge, the structural reduction $S(p) = \chi_p(r_- - r_+)$ and the resulting explicit evaluation of $S(p)$ in the full-rank regime $\alpha(p) = p - 1$ do not appear to have been recorded previously.

1.5. Organisation

Section 2 sets notation. Section 3 proves the primitivity result (Step I). Sections 4 and 5 carry out Step II. Section 6 executes Step III. Section 7 assembles the main theorem. Section 8 reports the computational verification, and Section 9 states explicit density and generalisation questions.

2. Preliminaries

Throughout, p denotes an odd prime $\neq 5$, with $p \geq 7$ unless explicitly stated otherwise. We use the standard extension of the quadratic character to all of $\mathbb{F}_p$ by setting $\chi_p(0) = 0$, and recall that $\chi_p(-1) = (-1)^{(p-1)/2}$. We write

$$\text{QR}_p^\times = \{u \in \mathbb{F}_p^\times : \chi_p(u) = +1\}, \quad \text{NQR}_p = \{u \in \mathbb{F}_p^\times : \chi_p(u) = -1\}.$$

When $p \equiv \pm 1 \pmod{5}$, the polynomial $x^2 - x - 1$ splits over $\mathbb{F}_p$. Fix a square root $s \in \mathbb{F}_p$ of 5, and write its two roots as

$$r_1 = \frac{1+s}{2}, \quad r_2 = \frac{1-s}{2}.$$

Then $r_1 + r_2 = 1$, $r_1 r_2 = -1$, and $r_1 - r_2 = s$. The Binet formula holds in $\mathbb{F}_p$:

$$F_m = \frac{r_1^m - r_2^m}{r_1 - r_2}. \quad (2)$$

After the quadratic-character analysis in Lemma 3.2, in the full-rank case we shall denote the nonresidue root by r_- and the residue root by r_+ , so that

$$\chi_p(r_-) = -1, \quad \chi_p(r_+) = +1, \quad r_- + r_+ = 1, \quad r_- r_+ = -1.$$

The Binet formula may then be written as $F_m = (r_-^m - r_+^m)/(r_- - r_+)$. The choice of s will additionally be fixed by $\chi_p(s) = +1$ in Section 6; this is independent of the labels $r_\pm$, which are determined by quadratic character.

Elementary group-theoretic facts

Lemma 2.1 (Quadratic character in a cyclic group). *Let g be a primitive root of $\mathbb{F}_p^\times$. Then $\chi_p(g^k) = (-1)^k$ for every integer k .*

Proof. The nonzero squares in $\mathbb{F}_p^\times$ are exactly the even powers of g . Since $p - 1$ is even, the parity of k is well-defined modulo $p - 1$. ■

Lemma 2.2 (Primitive-root reindexing). *Let g be a primitive root of $\mathbb{F}_p^\times$. The map $n \mapsto g^n$ from $\{1, \dots, p-1\}$ to $\mathbb{F}_p^\times$ is a bijection. Consequently, for any function $H : \mathbb{F}_p^\times \rightarrow A$ with A abelian,*

$$\sum_{n=1}^{p-1} H(g^n) = \sum_{u \in \mathbb{F}_p^\times} H(u).$$

Moreover, $\chi_p(g^n) = (-1)^n$ for $1 \leq n \leq p - 1$.

Proof. Since g has order $p - 1$, the powers $g, g^2, \dots, g^{p-1}$ are exactly the elements of $\mathbb{F}_p^\times$, each occurring once. The character formula follows from Lemma 2.1. ■

Lemma 2.3 (Fifth roots and the cyclotomic polynomial). *Let K be a field of characteristic $\neq 5$, and let $\Phi_5(X) = X^4 + X^3 + X^2 + X + 1$. For $z \in K$: $\Phi_5(z) = 0$ if and only if $z \in K^\times$, $z^5 = 1$, and $z \neq 1$. In particular, $\Phi_5(z) = 0$ implies $\text{ord}(z) = 5$.*

Proof. From $X^5 - 1 = (X - 1)\Phi_5(X)$: if $\Phi_5(z) = 0$ then $z^5 = 1$; and $\Phi_5(1) = 5 \neq 0$ gives $z \neq 1$. Conversely, $z^5 = 1$ and $z \neq 1$ force $(z - 1)\Phi_5(z) = 0$ and $z - 1 \neq 0$, so $\Phi_5(z) = 0$. ■

3. The structure lemmas (Step I)

Throughout this section, p denotes an odd prime with $p \neq 5$. Whenever roots are used as elements of $\mathbb{F}_p$, the splitting hypothesis will be stated explicitly.

Lemma 3.1 (Rank-order identity). *Let r_1, r_2 be the two roots of $x^2 - x - 1$ in $\mathbb{F}_{p^2}$, and put $q = r_1/r_2$. Then $\alpha(p) = \text{ord}_{\mathbb{F}_{p^2}}(q)$. In particular, if $x^2 - x - 1$ splits over $\mathbb{F}_p$, then for each root $r \in \mathbb{F}_p$, $\alpha(p) = \text{ord}_{\mathbb{F}_p^\times}(-r^2)$.*

Proof. Since $p \neq 5$, the roots are distinct. Binet's formula in $\mathbb{F}_{p^2}$ gives $F_m \equiv 0 \pmod{p} \iff r_1^m = r_2^m \iff q^m = 1$, so $\alpha(p) = \text{ord}(q)$. In the split case, fix a root $r \in \mathbb{F}_p$. The other root is $-r^{-1}$, since the product of the roots is -1 . Thus the ratio r_1/r_2 is either $-r^2$ or its inverse, depending on the ordering of r_1, r_2 . These two elements have the same multiplicative order. Hence

$$\alpha(p) = \text{ord}_{\mathbb{F}_p^\times}(-r^2).$$

■

Lemma 3.2 (Primitive root mechanism in the full-rank regime). *Let $p \geq 7$ be a prime such that $\alpha(p) = p - 1$. Then $x^2 - x - 1$ splits over $\mathbb{F}_p$, and $p \equiv 3 \pmod{4}$. Label its two roots $r_+, r_- \in \mathbb{F}_p^\times$ by their quadratic characters: $\chi_p(r_+) = +1$, $\chi_p(r_-) = -1$. Then*

$$\text{ord}_{\mathbb{F}_p^\times}(-r_-^2) = p - 1 \quad \text{and} \quad \text{ord}_{\mathbb{F}_p^\times}(r_-) = p - 1.$$

In particular, r_- is a primitive root of $\mathbb{F}_p^\times$.

Proof. Let r_1, r_2 be the two roots of $x^2 - x - 1$ in $\mathbb{F}_{p^2}$, and put $q = r_1/r_2$.

Splitting. Suppose $x^2 - x - 1$ does not split over $\mathbb{F}_p$. The Frobenius endomorphism $\varphi: x \mapsto x^p$ of $\mathbb{F}_{p^2}$ fixes $\mathbb{F}_p$ pointwise and interchanges the two roots of any irreducible quadratic over $\mathbb{F}_p$; hence $r_1^p = r_2$ and $r_2^p = r_1$. Thus

$$q^p = \frac{r_1^p}{r_2^p} = \frac{r_2}{r_1} = q^{-1},$$

giving $q^{p+1} = 1$ and $\text{ord}(q) \mid p + 1$. By Lemma 3.1, $\alpha(p) \mid p + 1$, so $p - 1 \mid p + 1$, hence $p - 1 \mid 2$, contradicting $p \geq 7$. Therefore $x^2 - x - 1$ splits over $\mathbb{F}_p$.

Congruence $p \equiv 3 \pmod{4}$. Let r be either root in $\mathbb{F}_p$. By the split-case part of Lemma 3.1, $\text{ord}(-r^2) = p - 1$. If $\chi_p(-1) = +1$, then $-r^2$ is a nonzero square and hence lies in a subgroup of $\mathbb{F}_p^\times$ of order $(p-1)/2 < p-1$, contradicting $\text{ord}(-r^2) = p - 1$. Hence $\chi_p(-1) = -1$, i.e. $p \equiv 3 \pmod{4}$.

Existence of $r_\pm$. Since $r_1 r_2 = -1$ and $\chi_p(-1) = -1$, we have $\chi_p(r_1) \chi_p(r_2) = -1$, so exactly one root is a residue and the other a nonresidue. Label them r_+ and r_- accordingly.

Order of $-r_-^2$. By Lemma 3.1 applied to r_- : $\text{ord}(-r_-^2) = \alpha(p) = p - 1$.

Primitivity of r_- . Since $\chi_p(r_-) = -1$, Euler's criterion gives $r_-^{(p-1)/2} = -1$, so $-1 \in \langle r_- \rangle$. Since also $r_-^2 \in \langle r_- \rangle$, we have $-r_-^2 \in \langle r_- \rangle$. But $\text{ord}(-r_-^2) = p - 1$, so the subgroup $\langle r_- \rangle$ contains an element of order $p - 1 = |\mathbb{F}_p^\times|$. Therefore $\langle r_- \rangle = \mathbb{F}_p^\times$ and $\text{ord}(r_-) = p - 1$. ■

Remark 3.3 (The content of Lemma 3.2). *The content of Lemma 3.2 is threefold: full rank forces $x^2 - x - 1$ to split over $\mathbb{F}_p$ (by a Frobenius argument), forces $p \equiv 3 \pmod{4}$, and lifts the primitivity of $-r_-^2$ to r_- itself via $-1 \in \langle r_- \rangle$. This is the structural fact that makes the substitution $u = r_-^n$ in Step II a bijection onto $\mathbb{F}_p^\times$, and it is the key structural input in the evaluation of $S(p)$.*

Remark 3.4 (Order of the residue root). *In the full-rank regime, put $n = (p - 1)/2$. Since $p \equiv 3 \pmod{4}$, n is odd, and since r_- is primitive, $r_-^n = -1$. From $r_- r_+ = -1$, we get*

$$r_+ = -r_-^{-1} = r_-^{n-1}.$$

Writing $p = 4m + 3$, so $n = 2m + 1$, gives

$$\text{ord}(r_+) = \frac{p-1}{\gcd(p-1, n-1)} = \frac{4m+2}{\gcd(4m+2, 2m)} = \frac{4m+2}{2} = \frac{p-1}{2}.$$

4. Character sum identities

Lemma 4.1. *For every $a \in \mathbb{F}_p^\times$, $\sum_{x \in \mathbb{F}_p} \chi_p(x^2 - a) = -1$.*

Proof. This is classical; see [9]. Put

$$T(a) := \sum_{x \in \mathbb{F}_p} \chi_p(x^2 - a).$$

Using $\#\{x : x^2 = t\} = 1 + \chi_p(t)$ and $\sum_t \chi_p(t - a) = 0$,

$$T(a) = \sum_{t \in \mathbb{F}_p} (1 + \chi_p(t)) \chi_p(t - a) = \sum_t \chi_p(t(t - a)).$$

The substitution $t = aw$ shows $T(a)$ is independent of $a \neq 0$. Since $T(0) = p - 1$ and $\sum_{a \in \mathbb{F}_p} T(a) = 0$, the common value is -1 . ■

Lemma 4.2. *If $p \equiv 3 \pmod{4}$, the map $u \mapsto -u$ is a bijection $\text{QR}_p^\times \rightarrow \text{NQR}_p$.*

Proof. $\chi_p(-u) = \chi_p(-1) \chi_p(u) = -\chi_p(u)$, and both sets have cardinality $(p-1)/2$. ■

Lemma 4.3. *If $p \equiv 3 \pmod{4}$, then*

$$A := \sum_{u \in \text{QR}_p^\times} \chi_p(u^2 - 1) = 0, \quad B := \sum_{u \in \text{NQR}_p} \chi_p(u^2 + 1) = -1.$$

Proof. By Lemma 4.1 with $a = 1$, isolating the $x = 0$ term (where $\chi_p(-1) = -1$):

$$\sum_{x \in \mathbb{F}_p^\times} \chi_p(x^2 - 1) = -1 - \chi_p(-1) = 0.$$

Since $(-u)^2 - 1 = u^2 - 1$, the involution $u \mapsto -u$ maps $\text{QR}_p^\times \rightarrow \text{NQR}_p$ bijectively (Lemma 4.2), giving

$$A = \sum_{u \in \text{QR}_p^\times} \chi_p(u^2 - 1) = \sum_{v \in \text{NQR}_p} \chi_p(v^2 - 1).$$

Hence $2A = \sum_{x \in \mathbb{F}_p^\times} \chi_p(x^2 - 1) = 0$, so $A = 0$.

By Lemma 4.1 with $a = -1$, isolating $x = 0$ (where $\chi_p(1) = +1$):

$$\sum_{x \in \mathbb{F}_p^\times} \chi_p(x^2 + 1) = -1 - 1 = -2.$$

Since $(-u)^2 + 1 = u^2 + 1$, the same involution gives $\sum_{u \in \text{NQR}_p} \chi_p(u^2 + 1) = \sum_{v \in \text{QR}_p^\times} \chi_p(v^2 + 1)$. Thus $2B = -2$ and $B = -1$. ■

5. The structural identity (Step II)

Proposition 5.1. *Let $p \geq 7$ with $\alpha(p) = p - 1$. Then*

$$S(p) = \chi_p(r_- - r_+),$$

where r_- and r_+ are the nonresidue and residue roots of $x^2 - x - 1$ in $\mathbb{F}_p$.

Proof. Let $D = r_- - r_+$. Since the discriminant of $x^2 - x - 1$ is $5 \neq 0$ in $\mathbb{F}_p$, we have $D \neq 0$. Since $r_+ = -r_-^{-1}$, Binet's formula gives

$$F_n = \frac{r_-^n - (-1)^n r_-^{-n}}{D}.$$

By Lemma 3.2, r_- is a primitive root of $\mathbb{F}_p^\times$. By Lemma 2.2, $n \mapsto u = r_-^n$ is a bijection $\{1, \dots, p-1\} \rightarrow \mathbb{F}_p^\times$ with $\chi_p(u) = (-1)^n$. Setting $\varepsilon := \chi_p(D)$:

$$\chi_p(F_n) = \varepsilon \cdot \chi_p(u - \chi_p(u)u^{-1}).$$

For $u \in \text{QR}_p^\times$ ($\chi_p(u) = +1$): contribution $\varepsilon \chi_p(u^2 - 1)$. For $u \in \text{NQR}_p$ ($\chi_p(u) = -1$): contribution $-\varepsilon \chi_p(u^2 + 1)$. By Lemma 4.3, $A = 0$ and $B = -1$, so

$$S(p) = \varepsilon(A - B) = \varepsilon(0 - (-1)) = \varepsilon = \chi_p(r_- - r_+). \quad \blacksquare$$

Remark 5.2. *The $n = p-1$ term (where $u = 1$, $u^2 - 1 = 0$) contributes 0 to A and does not affect the sum.*

6. Explicit sign via fifth roots of unity (Step III)

Assume $p \equiv 3 \pmod{4}$ and $p \equiv \pm 1 \pmod{5}$. Since $\chi_p(5) = 1$, choose $s \in \mathbb{F}_p$ with $s^2 = 5$ and $\chi_p(s) = +1$ (the two square roots of 5 have opposite characters since $\chi_p(-1) = -1$). Put

$$\phi = \frac{1+s}{2}, \quad \psi = \frac{1-s}{2}.$$

These are the two roots of $x^2 - x - 1$. The labels r_-, r_+ will be identified in Lemma 6.3.

Lemma 6.1. *There exists $z \in \mathbb{F}_p^\times$ with $z^5 = 1$ and $z \neq 1$ if and only if $p \equiv 1 \pmod{5}$.*

Proof. $\mathbb{F}_p^\times$ is cyclic of order $p-1$; it contains an element of order 5 iff $5 \mid p-1$. ■

Lemma 6.2 (Discriminant criterion). *Let $t = (s-1)/2$ and $\Delta = t^2 - 4$. Then $\Delta \neq 0$ and*

$$\chi_p(\Delta) = \begin{cases} +1, & p \equiv 1 \pmod{5}, \\ -1, & p \equiv -1 \pmod{5}. \end{cases}$$

Proof. Setup. From $s^2 = 5$, one computes $t^2 + t - 1 = 0$ and $\Delta = -t - 3$. If $\Delta = 0$ then $t = -3$, giving $5 = 0$ in $\mathbb{F}_p$, contradicting $p \neq 5$.

($\Rightarrow$): By the discriminant criterion, $\chi_p(\Delta) = +1$ iff $Z^2 - tZ + 1 = 0$ has a root $z \in \mathbb{F}_p$. Any such z satisfies $z + z^{-1} = t$, hence $z^2 + z^{-2} = t^2 - 2$, and therefore

$$(z^2 + z^{-2}) + (z + z^{-1}) + 1 = (t^2 - 2) + t + 1 = t^2 + t - 1 = 0.$$

Multiplying by z^2 gives $\Phi_5(z) = 0$; by Lemma 2.3, $\text{ord}(z) = 5$, so $p \equiv 1 \pmod{5}$ by Lemma 6.1.

($\Leftarrow$): Assume $p \equiv 1 \pmod{5}$. Choose $z_0 \in \mathbb{F}_p^\times$ of order 5 and set $\tau = z_0 + z_0^{-1}$. From $\Phi_5(z_0) = 0$ one deduces $\tau^2 + \tau - 1 = 0$. The two roots of $X^2 + X - 1$ over $\mathbb{F}_p$ are t and $-1 - t$.

If $\tau = t$, then z_0 satisfies $z_0^2 - tz_0 + 1 = 0$, giving $\chi_p(\Delta) = +1$.

If $\tau = -1 - t$, set $z_1 = z_0^2$ (still of order 5). Then $z_1 + z_1^{-1} = \tau^2 - 2 = 1 - \tau - 2 = -1 - \tau = t$, so z_1 satisfies $z_1^2 - tz_1 + 1 = 0$, again giving $\chi_p(\Delta) = +1$. Thus $p \equiv 1 \pmod{5}$ implies $\chi_p(\Delta) = +1$. Together with the implication proved above, this gives

$$\chi_p(\Delta) = +1 \iff p \equiv 1 \pmod{5}.$$

Since $\Delta \neq 0$ and throughout this section $p \equiv \pm 1 \pmod{5}$, the remaining case $p \equiv -1 \pmod{5}$ has $\chi_p(\Delta) = -1$. ■

Lemma 6.3 (Sign determination). *With r_- the nonresidue root and r_+ the residue root of $x^2 - x - 1$,*

$$\chi_p(r_- - r_+) = \begin{cases} +1, & p \equiv 1 \pmod{5}, \\ -1, & p \equiv -1 \pmod{5}. \end{cases}$$

Proof. Compute $s\phi = s(1+s)/2 = (s+5)/2$. By $\Delta = -t-3$ and $t = (s-1)/2$,

$$\Delta = -\frac{s-1}{2} - 3 = -\frac{s+5}{2} = -s\phi.$$

Taking characters (using $\chi_p(s) = +1$ and $\chi_p(-1) = -1$):

$$\chi_p(\Delta) = -\chi_p(\phi).$$

Case $p \equiv 1 \pmod{5}$. By Lemma 6.2, $\chi_p(\Delta) = +1$, so $\chi_p(\phi) = -1$. Hence $r_- = \phi$, $r_+ = \psi$, and $r_- - r_+ = \phi - \psi = s$, giving $\chi_p(r_- - r_+) = +1$.

Case $p \equiv -1 \pmod{5}$. By Lemma 6.2, $\chi_p(\Delta) = -1$, so $\chi_p(\phi) = +1$. Hence $r_+ = \phi$, $r_- = \psi$, and $r_- - r_+ = \psi - \phi = -s$, giving $\chi_p(r_- - r_+) = \chi_p(-1)\chi_p(s) = -1$. ■

7. Proof of the main theorem

Proof of Theorem 1.1. By Lemma 3.2, $x^2 - x - 1$ splits over $\mathbb{F}_p$ and $p \equiv 3 \pmod{4}$. Splitting is equivalent to $\chi_p(5) = +1$, hence $p \equiv \pm 1 \pmod{5}$. Combining gives $p \equiv 11 \pmod{20}$ or $p \equiv 19 \pmod{20}$.

By Proposition 5.1, $S(p) = \chi_p(r_- - r_+)$. By Lemma 6.3, this is $+1$ when $p \equiv 1 \pmod{5}$ (i.e. $p \equiv 11 \pmod{20}$) and -1 when $p \equiv -1 \pmod{5}$ (i.e. $p \equiv 19 \pmod{20}$). ■

8. Computational verification

Theorem 1.1 was verified for all primes $p \leq 2 \times 10^6$ satisfying $\alpha(p) = p - 1$.

Table 1: Verification of Theorem 1.1 for all primes $p \leq 2 \times 10^6$ with $\alpha(p) = p - 1$.

$p \pmod{20}$	predicted $S(p)$	count	matches	status
11	+1	11,755	11,755	PASS
19	-1	14,652	14,652	PASS
total		26,407	26,407	PASS

The verification is non-circular: $\alpha(p)$ is computed by iterating the Fibonacci recurrence modulo p , and $S(p)$ is computed by directly summing $\chi_p(F_n)$ for $1 \leq n \leq p - 1$, without using the closed formula of Theorem 1.1. Consistency checks independently verified: (i) the quadratic characters of both roots of $x^2 - x - 1$ in $\mathbb{F}_p$ against Lemma 3.2; (ii) the multiplicative orders of $-r_-^2$ and r_- against Lemmas 3.1 and 3.2; and (iii) the sign of $\chi_p(r_- - r_+)$ against the discriminant criterion of Lemma 6.3. These checks verify not only the numerical value of $S(p)$ but also the structural mechanism used in the proof. Among the 148,933 primes $p \leq 2 \times 10^6$, exactly 26,407 satisfy $\alpha(p) = p - 1$. Restricting to the 74,461 split primes ($p \equiv \pm 1 \pmod{5}$), the full-rank density is $26,407/74,461 \approx 35.46\%$. Source code, datasets, and verification logs are deposited in the reproducibility archive [16].

9. Concluding remarks

We have established an explicit evaluation of the Fibonacci character sum $S(p)$ in the full-rank regime $\alpha(p) = p - 1$, showing that $S(p) = \pm 1$ with sign determined by $p \pmod{20}$. The split-case identity in Lemma 3.1 identifies $\alpha(p) = p - 1$ with the primitivity of $-r^2 \in \mathbb{F}_p^\times$, placing the result naturally in the Artin-type family.

A density question

Among primes $p \leq 2 \times 10^6$ in the split classes $p \equiv \pm 1 \pmod{5}$, approximately 35.46% satisfy $\alpha(p) = p - 1$. Theorem 1.1 restricts these primes to the two classes $p \equiv 11, 19 \pmod{20}$, out of the four equidistributed split classes $p \equiv 1, 9, 11, 19 \pmod{20}$. This motivates the following.

Conjecture 9.1. *There exists a constant C_F with $0 < C_F \leq 1/2$ such that*

$$\#\{p \leq x : \alpha(p) = p - 1\} \sim C_F \cdot \#\{p \leq x : p \equiv \pm 1 \pmod{5}\}, \quad x \rightarrow \infty$$

Equivalently, the natural density of full-rank primes among all primes equals $C_F/2$.

The upper bound $C_F \leq 1/2$ is forced by Theorem 1.1: full-rank primes can occur only in the residue classes 11 and 19 $\pmod{20}$, among the four split residue classes 1, 9, 11, 19 $\pmod{20}$. By the split-case identity in Lemma 3.1, Conjecture 9.1 is a primitive-root statement for the reduction of the algebraic unit $-\phi^2$ (where $\phi = (1 + \sqrt{5})/2$) at primes split in $\mathbb{Q}(\sqrt{5})$. A Hooley-type analysis in this conditional setting, under suitable GRH-type hypotheses [15, 14], may lead to an explicit Euler-product candidate for C_F . Establishing such a conditional formula lies beyond the scope of this paper.

Further directions

The methods apply directly to Lucas-type sums via $L_n = r_-^n + r_+^n$. The analysis of $S(p)$ in non-extremal regimes and higher-order character sums are left to future work.

Appendix A. Illustrative examples

For $p = 11$: $\sqrt{5} \equiv 4 \pmod{11}$; roots $r_- = 8, r_+ = 4$; $r_- - r_+ = 4 = 2^2$; $S(11) = +1$. For $p = 19$: roots $r_- = 15, r_+ = 5$; $r_- - r_+ = 10$ (nonresidue); $S(19) = -1$. For $p = 31$: $\alpha(31) = 30$; roots $r_- = 13, r_+ = 19$; $r_- - r_+ \equiv 25 = 5^2 \pmod{31}$; $S(31) = +1$. Also: $\alpha(59) = 58$, $S(59) = -1$; $\alpha(79) = 78$, $S(79) = -1$. All consistent with Theorem 1.1.

Appendix B. Reproducibility archive

Source code, datasets, and verification logs are deposited at [16].

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